

Assume that the concentration of ion x^+ in the extracellular fluid is 300 mEq/lit and inside the cell is 20 mEq/lit, the resting membrane potential is closer to

$\log_{10}(15) = 1.17$ $\log_{10}(0.06) = -1.2$

+71 mv

-73 mv

+94 mv

-61 mv

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Comparing Goldman equation with Nernst equation, which of the following is correct?

The permeability of the membrane to ions is only considered in the Nernst equation.

The Nernst equation shows the simultaneous effect of several ions on the membrane potential.

In the Nernst equation, only the concentration gradient of positively charged ions is included in the formula.

Goldman equation describes real conditions, while Nernst equation depicts imaginary conditions.

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Which of the following does not play any role in determining the level of resting membrane potential?

Leakage of K through the K leak channels

Leakage of Na through the K leak channels

Leakage of Na through the Na leak channels

Na-K Pump

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The peak of an action potential in a large nerve cell is closer to the Nernst potential of which of the following ions?

K

Na

Cl

HCO₃

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What is the main cause of repolarization phase of an action potential?

Closure of voltage gated K channels

Opening of voltage gated K channels

Opening of voltage gated Ca channels

Closure of voltage gated Na channels

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A new AP cannot occur in an excitable fiber as long as the membrane is still depolarized from the preceding action potential. Why?

Because of inactivation of voltage gated Na channels

Because of inactivation of voltage gated K channels

Because of activation of voltage gated Na channels

Because of activation of voltage gated K channels

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Salutatory conduction is of value for	الف	۷
decrease in conduction velocity	ب	*
conserving energy	ج	
having plateau phase	د	
existence of schwann cells		

In plateau phase, calcium is transported inside the cardiac cell by	الف	۸
Na^+/Ca^{2+} -exchanger	ب	
Ca^{2+} -ATPase	ج	
Na^+/K^+ -ATPase pump	د	*
voltage gated ca channels		

In which compartment is the largest amount of body water found?	الف	۹
ECF	ب	*
ICF	ج	
Plasma	د	
Interstitial fluid		

Which of the following acts as β cell glucose sensor?	الف	۱۰
GLUT 1	ب	*
GLUT 2	ج	
GLUT 3	د	
GLUT 4		

Which of the following transporters has the most number in the kidneys?	الف	۱۱
SGLT 1	ب	*
SGLT 2	ج	
SGLT 3	د	
SGLT 4		

Simple diffusion of ions is done based on:	الف	۱۲
chemical gradient	ب	
concentration gradient	ج	
electrical gradient	د	*
electrochemical gradient		

Which of the following is negatively correlated with the rate of simple diffusion?	الف	۱۳
Lipid solubility	ب	*
Thickness of membrane	ج	
Concentration gradient across membrane	د	
Surface area of membrane		

Which of the following is the most important factor limiting the lateral movement of proteins in the membrane of epithelial cells in the kidney?	الف	۱۴
Connecting proteins to the cell cortex	ب	
Connecting proteins to the extracellular matrix	ج	
Connecting the proteins of one cell to the proteins of the adjacent cell	د	*
The existence of tight junctions between cells		

1	In which of the following cases, the existing protein is related to the relevant organelle?
*	1 Proteasome-ubiquitin
	2 Peroxisome-purine
	3 RER-clathrin
	4 Mitochondria - catalase

2	During the steps of protein synthesis in the RER, which of the following is not observed in the structure of this organelle?
	1 Ribosome receptor
	2 Translocator complex
	3 Signal Recognition Particle receptor
*	4 Signal peptide

3	Microfilaments are composed of protein and play a role in
	1 Vimentin- endocytosis
	2 Tubulin- cilia formation
*	3 Actin - division of mitosis
	4 Desmin - mitochondrial function

4	Pale central nucleus, large and euchromatin, shows:
*	1 This cell is active in protein synthesis
	2 This cell starts apoptosis
	3 This cell is differentiated
	4 This cell is dead and necrotic

5	What are the names of the oligosaccharide chains that extend outward from the cell surface and what is their important function?
	1 Adherent glycoproteins - receptor
*	2 Glycocalyx - cell identification
	3 Basal layer - active transport
	4 Glycosaminoglycans - adhesion

6	In the Golgi organelle.....
	1 Cisternae are tubular
	2 Secretory vesicles are in cis face
*	3 cis cisternae are adjacent to the RER
	4 Golgi is located in the vicinity of the nucleus

7	Mitochondria organelle has which of the following characteristics?
	1 The outer membrane has thermogenin.
	2 The inner membrane has purine particles.
*	3 The cristae of the inner membrane enclose the mitochondrial matrix.
	4 The inner membrane contains ATPase

۸	Nucleosome has which of the following characteristics?	
	1	It has a core of 4 histones.
	2	It has single-stranded DNA
	3	It does not play a role in the formation of heterochromatin
	4	Histone H1 is outside the nucleosome.

۹	In receptor-mediated endocytosis	
	1	Clathrin is present in the membrane structure.
	2	Binding of ligand to the receptor causes the activity of ion channels
	3	Clathrin forms the coated pit at a faster rate
	4	Coated secretory vesicles have clathrin.

1	Obese people tend to have a _____ percentage of body water than thin people and females tend to have a _____ percentage than males
1	lower/higher
2	higher/ higher
*	3 lower/ lower
	4 higher/lower

2	All of the followings are major buffer systems that are responsible for regulating pH, <u>except</u> :
1	The phosphate
*	2 Sulfuric acid
	3 The bicarbonate
	4 Hemoglobin

3	Molecules that can be dissolved in nonpolar solvents are:
1	Amphipathic
2	Hydrophilic
3	Hypotonic
*	4 Hydrophobic

4	What is the pH of a solution whose OH^- is $1 \times 10^{-4} \text{ M}$
1	4
*	2 10
	3 6
	4 8

5	All of the following are considered as a function of proteins, <u>except</u> :
1	Antibodies
2	Catalysts
*	3 Insulation
	4 Transporter

6	Which of the following bonds considered strong chemical bonds individually?
1	Ionic bond/Hydrophobic bond
2	Hydrophobic bond /Covalent bonds
3	Ionic bond/Hydrogen bond
*	4 Disulfide bond/ Ionic bond

7	According to the pK value, which acid is stronger?
*	1 Acid 1: $\text{pK} = -10.3$
	2 Acid 2: $\text{pK} = -8.7$
	3 Acid 3: $\text{pK} = -6.5$
	4 Acid 4: $\text{pK} = -9.2$

8	Which of the following is monomer?
1	Protein
*	2 Vitamin
	3 Nucleic acid
	4 Polysaccharide

9	All statements are true about galactose with the exception of:
1	It is used for biosynthesis milk in mammary gland
2	It is a constituent for glycolipids
*	3 It is a ketoheptose
	4 The accumulation of galactose can lead to cataract

10	Choose an Aldopentose:
1	Glyceraldehyde
2	Dihydroxyacetone
3	Erythrose
*	4 Ribose

11	Which of the following monosaccharides is the precursor for synthesis of glycogen as a storage carbohydrate in animals?
1	Fructose
*	2 Glucose
	3 Ribose
	4 Mannose

12	Which of the following are enantiomers:
1	D-Galactose and L-Glucose
2	d- Galactose and I-Glucose
*	3 D-Mannose and L-Mannose
	4 α -Mannose and β -Mannose

13	Which statement is true about Polysaccharides?
1	Heteropolysaccharides contain only one type of monosaccharide unit
*	2 Glycogen is a homopolysaccharide
	3 All polysaccharides are linear
	4 Chondroitin sulfate contain sugar alcohol

14	A researcher wants to determine the glomerular filtration rate (GFR). Which of the following carbohydrates can be suggested?
*	1 Inulin
	2 Amylopectin
	3 Hyaluronic acid
	4 Amylose

15	Which biomolecule can be an appropriate candidate for control of calcification in bone?
1	Hyaluronic acid
2	Keratan sulphate
*	3 Chondrotin sulphate
	4 Dermatan sulphate

16	Maltose is a disaccharide of:
1	Fructose and lactose
*	2 Glucose and glucose
	3 Glucose and galactose
	4 Glucose and lactose

17	The slope of the Lineweaver-Burk Plot is:
1	V_{max}/K_m
*	2 K_m/V_{max}
	3 $V_{max}/[S]$
	4 $[S]/K_m$

18	Which of the following enzymes is an example of hydrolases?
*	1 Phosphatase
	2 Dehydrogenase
	3 Mutase
	4 Phosphorylase

19	Which of the following is the definition of enzyme activity?
1	The amount of the enzyme that catalyzes the conversion of one micromole of substrate per second.
2	The number of molecules of substrate converted to product per enzyme molecule per second.
3	The number of molecules of substrate converted to product per enzyme molecule per minute.
*	4 The amount of the enzyme that catalyzes the conversion of one micromole of substrate per minute.

20	Which of the following reaction is catalyzed by Lyase?
*	1 Breaking of bonds
	2 Intramolecular rearrangement of bonds
	3 Formation of bonds
	4 Transfer of group from one molecule to another

21	All of the following are the examples for competitive inhibitor, <u>except</u> ?
1	Methotrexate for dihydrofolate reductase
2	Atorvastatin for HMG-CoA reductase
*	3 Amoxicillin for transpeptidase
	4 Malonate for succinate dehydrogenase

22	Which of the following is <u>not</u> true regarding allosteric enzyme?
1	Activator increases the activity of an enzyme
2	Inhibitor decreases the activity after binding
3	Are generally multi-subunit enzymes
*	4 Display a hyperbolic dependence on substrate concentration

23	Reversible non-competitive inhibitor binds reversibly at a site other than the active site. This binding lead to:
1	Increase in K_m and V_{max} remains constant
*	2 Decrease in V_{max} and K_m remains constant
	3 Increase in V_{max} and K_m remains constant
	4 No change in V_{max} and K_m

24	An enzyme extract when subjected to electric field separates into two fractions, each catalyzing the same reaction. The fractions are:
1	Coenzyme
2	Allosteric enzyme
*	3 Isoenzyme
	4 Inducible enzyme

25	Which out of the following amino acids does not consider as a site of attachment for a phosphate group?
1	Serine
2	Threonine
*	3 Leucine
	4 Tyrosine

26	Hydroxyl containing amino acids are:
1	Cysteine and methionine
2	Threonine and cysteine
3	Serine and methionine
*	4 Serine and threonine

27	Which of the characteristics below apply to amino acid Glycine?
*	1 Optically inactive
	2 Hydrophilic, basic and charged
	3 Hydrophilic, acidic and charged
	4 Proton donor

28	Absorbance at 280 nm (UV wavelength) exhibited by protein is due to:
1	Aliphatic amino acids
*	2 Aromatic amino acids
	3 Acidic amino acids
	4 Basic amino acids

29	All of the following statement about Myoglobin is correct, except:
1	Consist of a single polypeptide chain
*	2 Its secondary structure is rich in β -sheet.
	3 Is a compact and spherical molecule
	4 Present in the skeletal muscle

30	What are the most common amino acids in collagen?
1	Tryptophan and Proline
	2 Glycine and Lysine
	3 Threonine and Proline
*	4 Proline and Glycine

31	Which of the followings increase P_{50} in the oxygen-dissociation curve of Hemoglobin?
1	Decrease of CO_2 pressure
	2 Increase of pH
	3 Decrease of temperature
*	4 Increase of 2,3 BPG

32	Which of the following proteins is defective in Ehlers -Danlos syndrome?
1	Myoglobin
	2 Elastase
*	3 Collagen
	4 Hemoglobin

33	Which one is correctly matched?
1	NADH-Vitamin B2
	2 $FADH_2$ -Vitamin B3
*	3 PLP-Vitamin B6
	4 TPP-Vitamin C

34	Thiamine deficiency causes all, except:
*	1 Cheilosis
	2 Ataxia
	3 Nystagmus
	4 Ophthalmoplegia

35	Which of the following diseases is caused by the deficiency of Niacin?
1	Scurvy
2	Rickets
*	3 Pellagra
	4 Pernicious anemia

36	Which of the following is effective in premenstrual syndrome (PMS) and diabetes?
1	Vitamin B12
2	Vitamin B9
*	3 Vitamin B6
	4 Vitamin B3

37	Which vitamin acts as a coenzyme for called γ -carboxylation reaction?
1	Vitamin A
*	2 Vitamin K
	3 Vitamin C
	4 Vitamin B7

38	Which nutrient deficiency causes megaloblastic anemia?
1	Pantothenic acid
2	Niacin
*	3 Folic acid
	4 Retinol

39	The most biologically active form of vitamin K is:
1	Menadione
*	2 Menaquinone
	3 Phylloquinone
	4 Phytonadion

40	Which vitamin is a major lipid-soluble antioxidant in cell membranes?
1	Vitamin A
2	Vitamin D
*	3 Vitamin E
	4 Vitamin K

41	Which of the following is a common feature of all phospholipids?
	1 A glycerol backbone
*	2 A phosphate group
	3 Two fatty acid tails
	4 A sphingosine backbone
42	Which lipid is a precursor for steroid hormone synthesis?
	1 Sphingomyelin
	2 Phosphatidylserine
*	3 Cholesterol
	4 Triacylglycerol
43	The primary function of HDL (high-density lipoprotein) is:
	1 Transporting cholesterol to peripheral tissues
	2 Transporting triglycerides to the liver
*	3 Transporting cholesterol from peripheral tissues to the liver
	4 Storing excess cholesterol in adipose tissue
44	In addition to its role in membrane structure, cholesterol is involved in:
	1 Energy storage
	2 Signal transduction
	3 Nerve impulse transmission
*	4 Regulation of membrane fluidity
45	A fatty acid which is not synthesized in the body and has to be supplied in the diet is
	1 Palmitic acid
	2 Lauric acid
*	3 Linolenic acid
	4 Palmitoleic acid
46	Cephalin consists of
	1 Glycerol, fatty acids, phosphoric acid and choline
*	2 Glycerol, fatty acids, phosphoric acid and ethanolamine
	3 sphingosine, 2 fatty acids, phosphoric acid and Inositol
	4 sphingosine, fatty acids, and serine

47	Glycosphingolipids are a combination of	
*	1	Ceramide with one or more sugar residues
	2	Glycerol with galactose
	3	Sphingosine with galactos
	4	Sphingosine with phosphoric acid

48	All the following have 18 carbon atoms <u>except</u> :	
	1	Linoleic acid
	2	Linolenic acid
*	3	Arachidonic acid
	4	Stearic acid